

Analog Electronics — Daily Schedule Term 4

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See also [Daily Schedule-Term 5](#)

Week 1 — Voltage, Current, Energy, Resistance

- Tuesday, Jan. 13
 - Theory: **What is Voltage?**
 - Practice: Radio Shack Learning Lab kit (hereafter just “kit”) Inventory
- Friday, Jan. 16
 - Practice: Load the batteries into the kit and separate out the resistors
 - Theory: **What is Resistance?**
 - Practice: Resistor color coding
 - Practice: Build the circuit diagrammed in **The Simplest Possible Circuit**

Week 2 — Power — Circuit Diagrams — Kirchhoff's Laws

- Tuesday, Jan. 20
 - Practice: The circuit diagram of **The Simplest Possible Circuit** that you built
 - Theory: **Resistance and Power, with Examples**
 - Practice: Build the two second-simplest circuits! You'll be reading circuit diagrams!
 - Build the circuit diagrammed in **Two Resistors in Series**
 - Build the circuit diagrammed in **Two Resistors in Parallel**
- Friday, Jan. 23
 - Theory: Come with the 20 problems in the last handout completed!
 - Practice: A circuit that lights an LED solidly
 - Theory: **Kirchhoff's Laws**
 - Practice: Let's build Mims' first circuit on pp. 14-18, a circuit that flashes an LED

Week 3 — Resistors in Series and Parallel — Voltage Dividers

- Tuesday, Jan. 27
 - Theory: Series and parallel resistors examples from Howard Gerrish, *Electronics*
 - Theory: How does mileage add? Example: fleet of two vehicles with 15mpg and 30mpg.
 - Practice: Build as much as we can on pp. 19-22, first a black box test circuit, and then a series of circuits that illustrate switches
 - Theory: **Resistors in Series and In Parallel** (write-up only has the circuit diagrams)
- Friday, Jan. 30
 - Theory: **Voltage Dividers and Capacitors**

Week 4 — Switches of Various Kinds — Switches in Series and in Parallel — Resistors in Practice — Building a Voltmeter

- Tuesday, Feb. 3
 - Practice: We had a ton of theory on Friday. Today, we'll just do practice using pp. 20-21 of Mims Workbook I
- Friday, Feb. 6
 - Theory: We started on **10 Practice Problems**
 - More Practice: We did pp. 22-25 of Mims Workbook I, which covered relays and repurposing an ammeter as a voltmeter

Week 5 — Voltage Dividers — Capacitors in Series and in Parallel

- Tuesday, Feb. 10
 - Theory: Go over the **10 Practice Problems** handed out on Friday — Derive (on blackboard) formulas for capacitors in series and parallel
 - Practice: Resume at p. 26 of Mims Workbook I
 - Homework for Friday: Come into lab on your own time and do pp. 27-28 of Mims
- Friday, Feb. 13
 - Homework: Any questions on pp. 27-28?
 - Theory: Discussion of the Wheatstone Bridge — Introduction to Alternating Current — **Capacitors and Alternating Current**
 - Practice: We'll resume on p. 29 of Mims, then skip pp. 30-31 and continue on p. 32-33 (capacitors)
 - Homework for Monday: Come into lab on your own time, read p. 32, and then do pp. 34-36 and 38 (note I am deliberately having you skip p. 37 which uses the black-box test circuit)

Week 6 — RC Circuits in Practice — Transformers and Inductors — Diodes

- Tuesday, Feb. 17
 - Homework: Any questions on the homework, pp. 32, 34-36 and 38?
 - Theory: Continue discussion of alternating current and capacitors in switched circuits — Also, capacitors interacting with relays
 - Practice: Resume at p. 39 of Mims Workbook I
 - Homework for Friday: p. 40 and 42-44 of Mims Workbook I
- Friday, Feb. 20
 - Homework: Any questions on the homework, pp. 40 and 42-44?
 - Theory: **Inductors and Transformers**
 - Practice: Resume at p. 45 of Mims
 - Skipping: In the interest of getting to p. 54 for Monday, we will be skipping pp. 48-53 (lots and lots of LED circuits)
 - Homework for Tuesday: pp. 46-47 and pp. 54-55 of Mims (your first transistor circuits)

Week 7 — Transistors!

- Tuesday, Feb. 24
 - Practice: Compare and contrast the transistor circuits on pp. 54-55
 - Theory: Insulators, Conductors, Semiconductors, and the Behavior of NPN transistors
 - Practical homework for Friday: Build the two-transistor circuits on pp. 56-57
 - Theory homework for Friday: Please get a start looking at **How Semiconductors Work** which we will discuss in the Museum on Friday
- Friday, Feb. 27
 - Practical homework: Would it be profitable to discuss and further understand the two two-transistor circuits you built on pp. 56-57?
 - Theory homework: Discuss **How Semiconductors Work**
 - Practice: We will skip p. 58, unless someone actually has a portable AM radio, and go straight to pp. 59-60 for our in-class practice
 - Tuesday, March 17 — Nothing new in Mims, but review the transistor circuits up to p. 60, review **How Semiconductors Work** and read the early history of **Silicon Valley and The Traitorous Eight**

HAVE A GREAT SPRING BREAK — SATURDAY, FEBRUARY 28 TO SUNDAY, MARCH 15